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Airbnb Price Prediction in London

Text Mining for Economics & Finance

4 March 2026

Motivation
Airbnb Pricing

Airbnb listing prices reflect observable housing characteristics captured by structured variables

Some quality signals may not be fully captured by structured variables

Listing descriptions may contain additional information about these signals

Research Question: Does listing text improve price prediction beyond structured housing features?

Population & Quantity of Interest

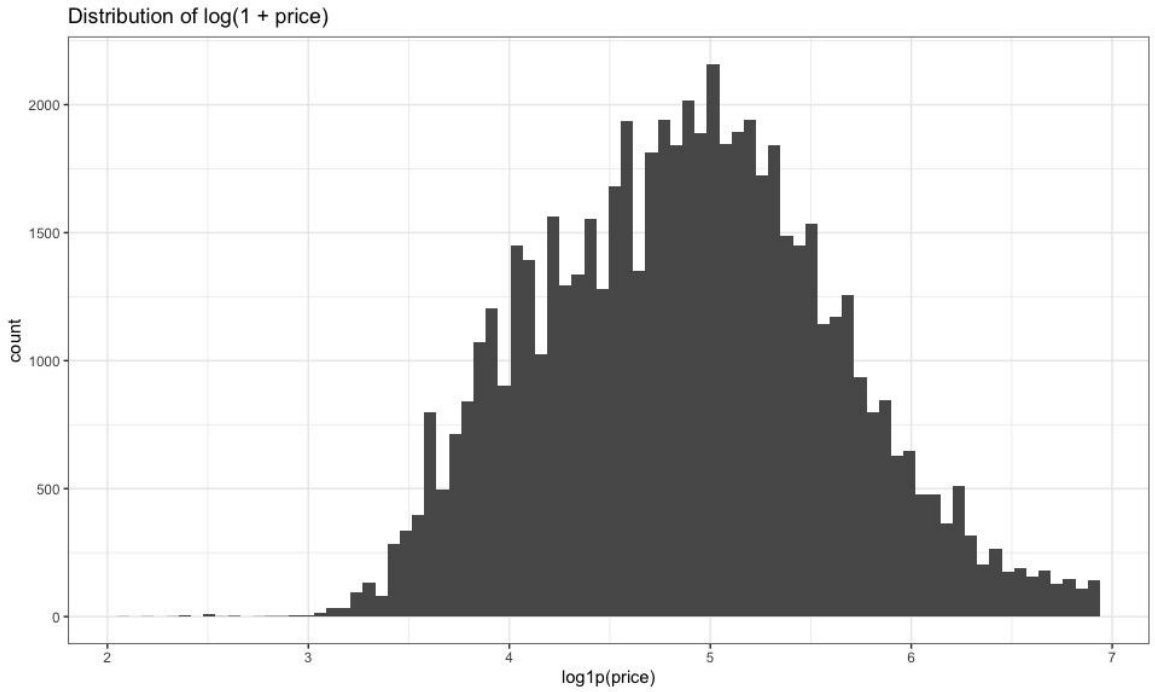
Airbnb Pricing

Population of Interest:

- London Airbnb Listings priced below £1000 per night
- We restrict ultra-luxury outliers to avoid a separate pricing regime.

Quantity of Interest

- Primary:
  - Log(price)
  - Log transformation stabilizes variance and improves model fit + interpretability
- Secondary:
  - Incremental reduction in prediction error when text is added



We study: How much additional price variation is explained by text?

Economics: Hedonic Pricing Framework

Airbnb Pricing

Economic intuition:
In a hedonic model, price reflects the bundle of characteristics a listing contains.

$$\log(price_i) = \beta_s X_i^{structured} + \beta_t X_i^{text} + \varepsilon_i$$

Where:

- $X^{structured}$: bedrooms, bathrooms, location, capacity
- X^{text}: linguistic signals from titles and descriptions

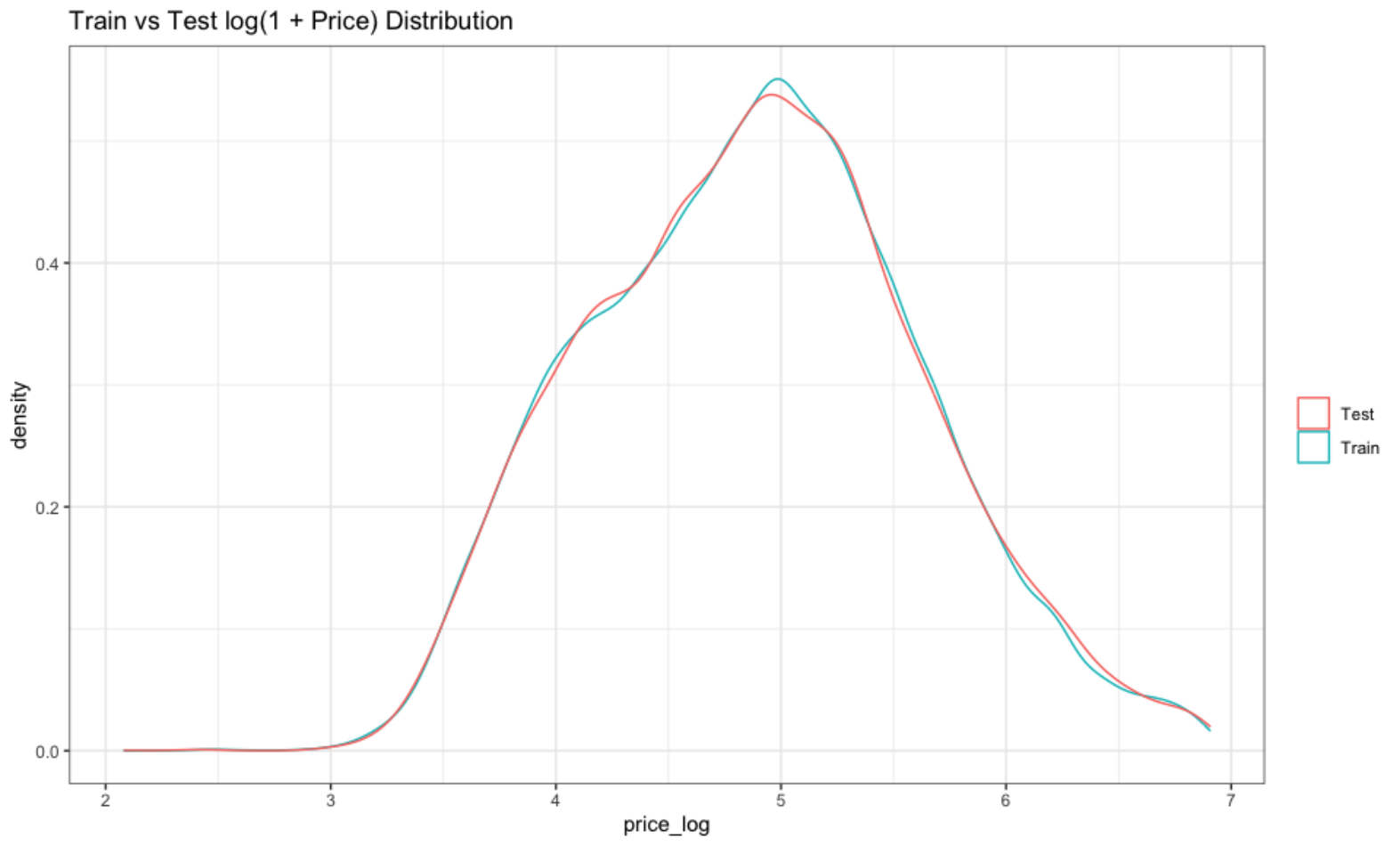
Key question:

Is β_t economically meaningful?

Data Overview

Airbnb Pricing

- ~60k listings
- Avg title length: 6 words
- Avg description length: 64 words
- Structured features:
  - latitude
  - longitude
  - bedrooms
  - bathrooms
  - beds
  - accommodates
- Train/Test split: 80/20 (stratified on log price)



Methods
Airbnb Pricing

Methods:

- N-grams (Bag-of-Words)
- Regularization
- Word Embeddings
- Transfer Learning
- Benchmark Comparisons

Benchmark Models

Airbnb Pricing

We must compare against simple baselines.

Benchmarks:

- Structured-only LASSO
- Description n-grams
- Embeddings-only

This shows whether complex NLP adds value.

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Section I: Title vs. Description

Which linguistic object better captures latent characteristics?

Title Models
Airbnb Pricing

1:2 n-grams

RMSE_log = 0.591

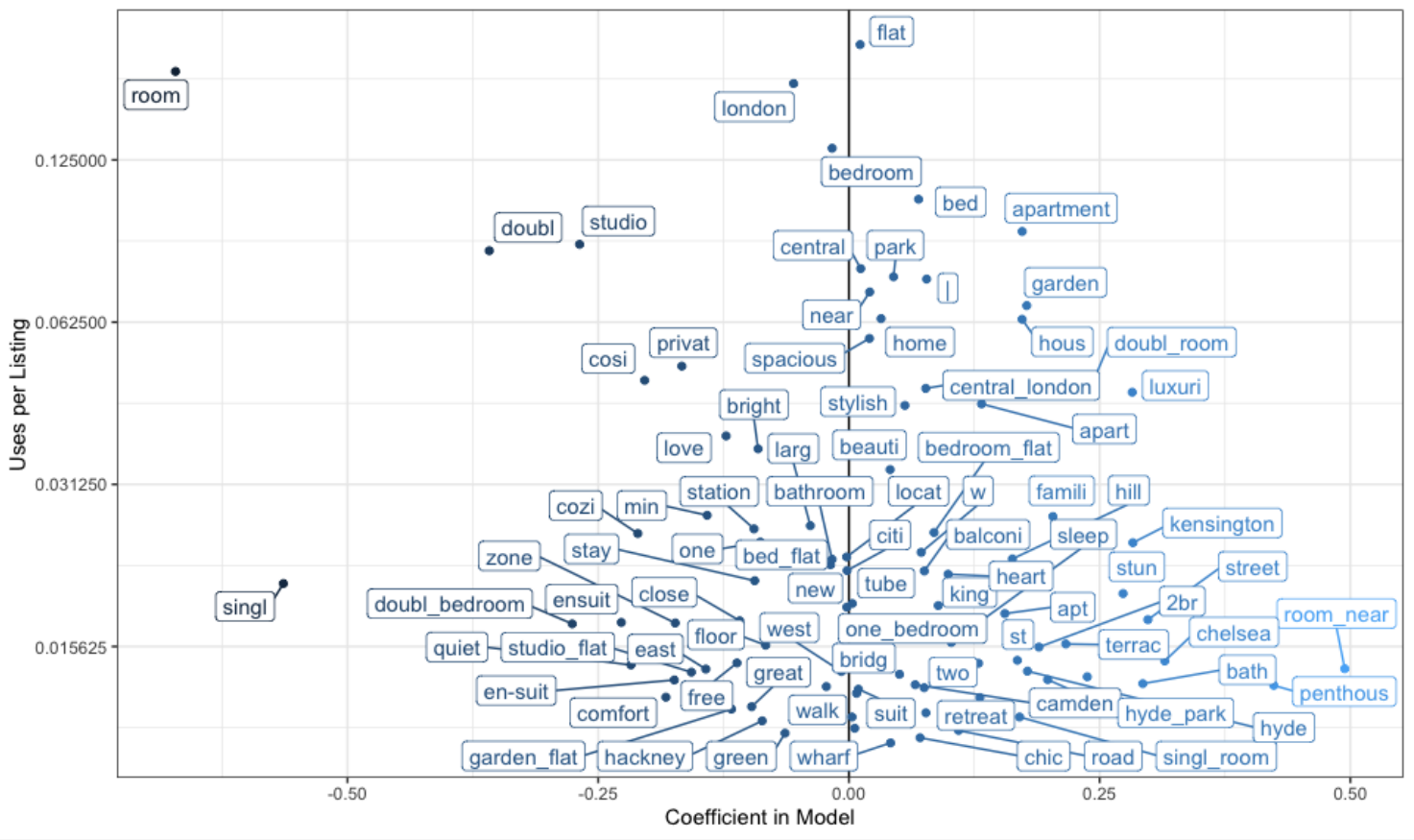
Interpretation:

Titles encode location & property type

But short documents → limited richness

Lesson:

Document length affects predictive power



Description n-Grams

Airbnb Pricing

Tested:

- Unigrams
- 1:2
- 1:2:3

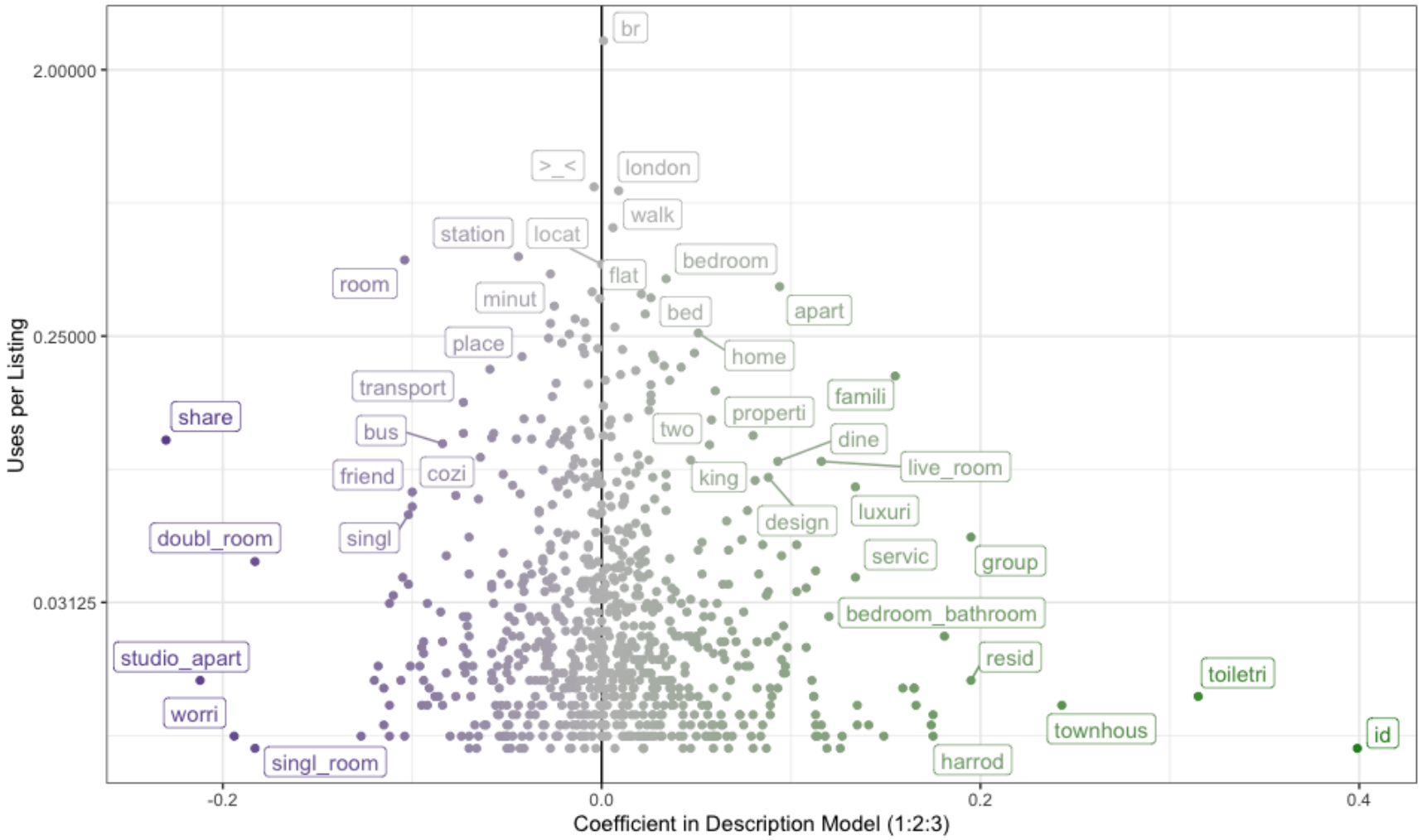
Best: 1:2:3

RMSE_log = 0.557

Interpretation:

Descriptions contain structured economic signals

- Location prestige
- Accommodation type
- Amenities



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Section II: Embeddings

Are economically meaningful signals embededed in semantic structure, not just word presence?

Transfer Learning (Embeddings)

Airbnb Pricing

Instead of estimating word meaning from scratch:
We use **pre-trained word embeddings**.

What is Transfer Learning?

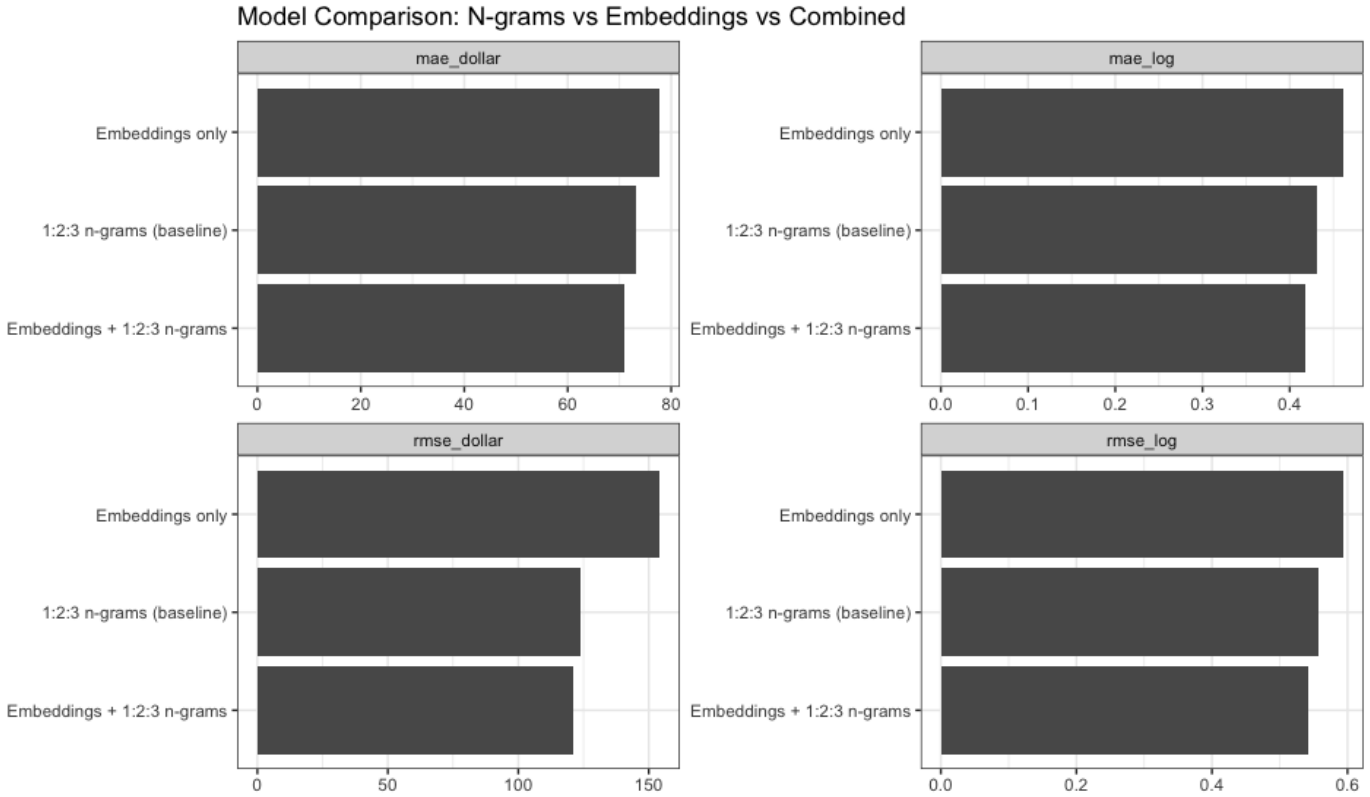
- Vectors trained on large external corpora
- Capture semantic similarity
- Project Airbnb descriptions into 300-dim space

Embeddings only: RMSE_log = 0.593

Embeddings + n-grams: RMSE_log = 0.543

Insight:

Transfer learning improves semantic richness,
but embeddings alone underperform sparse n-grams.



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Section III: Structures vs Structured + Text

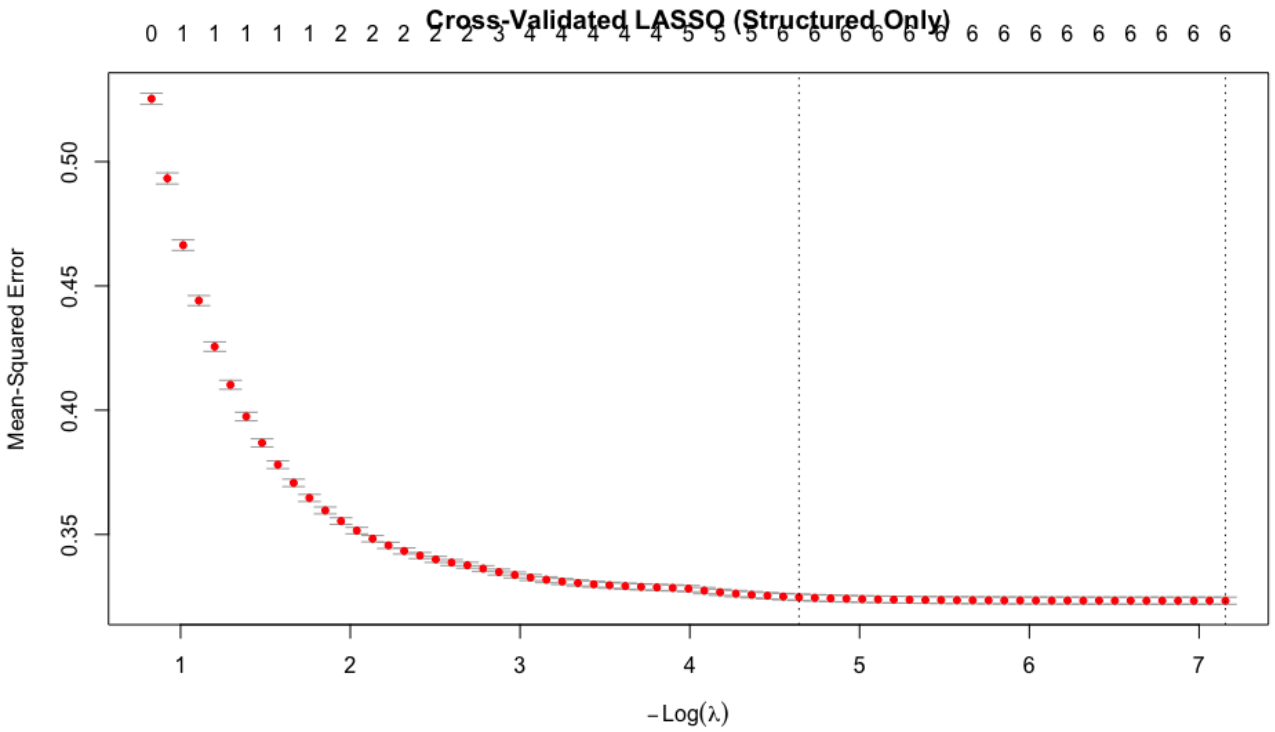
Does description text provide incremental explanatory power beyond structured housing fundamentals in the hedonic pricing function?

Structured Model
Airbnb Pricing

Features (Structured Fundamentals):

- Bedrooms
- Bathrooms
- Capacity
- Geographic coordinates

RMSE_log = 0.570



Interpretation:

Housing fundamentals explain most variation.

Text alone predicts price about as well as structured fundamentals.

Structured + Text (Best Model)

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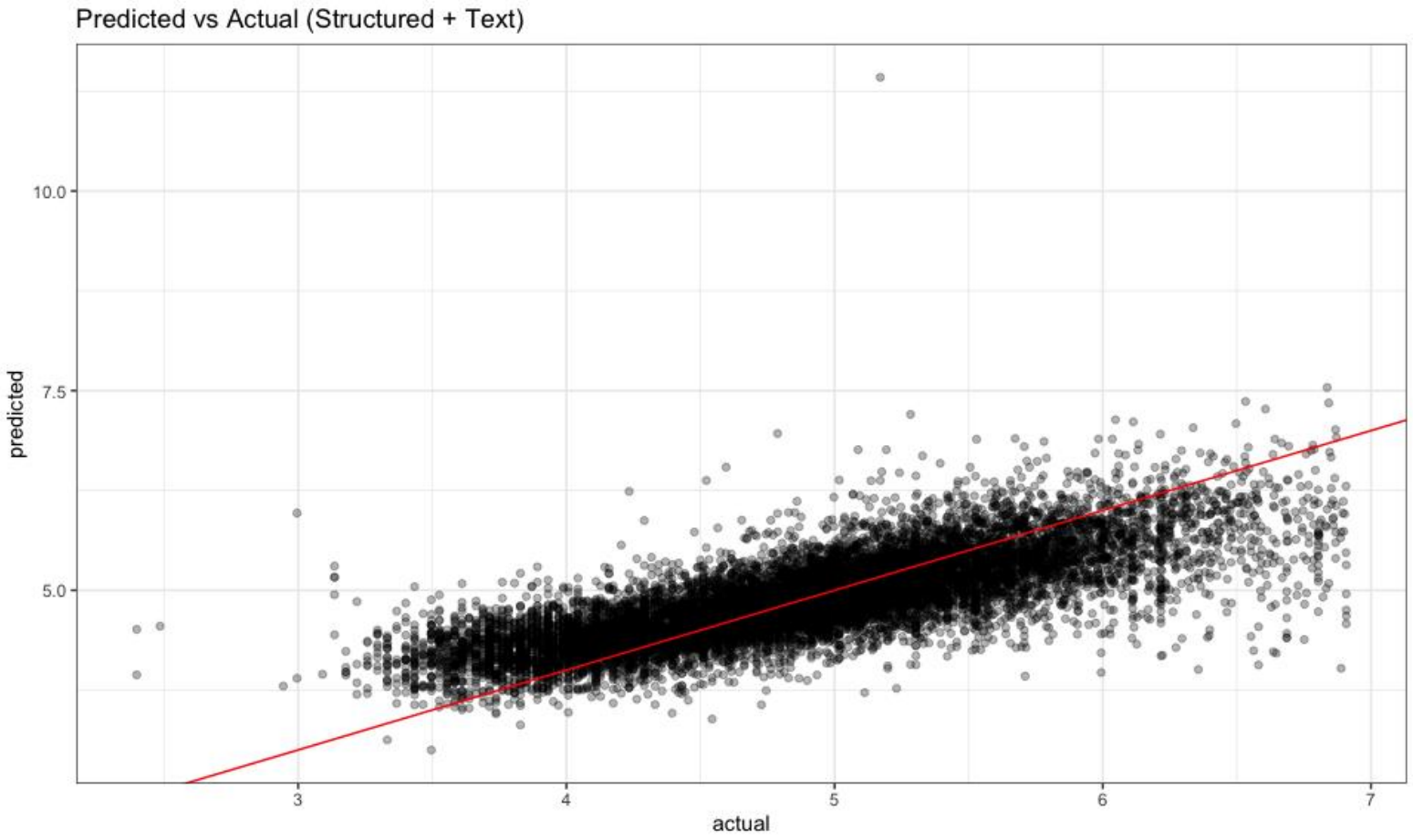
Combined:

- Structured
- Embeddings (transfer learning)
- 1:2:3 n-grams

RMSE_log = 0.464

This is the best performing model.



Interpretation:

Both capture different aspects of listing

- Structured: housing fundamentals
- Description: additional signals (amenities, prestige)

Model Comparison

Airbnb Pricing

Model	RMSE_log
Structured Only	0.570
Text (1:2:3)	0.557
Embeddings + Text	0.543
Structured + Text	**0.464**

Key Economic Result
Text reduces prediction error substantially when combined with structure.

This implies:
Descriptions encode incremental economic information.

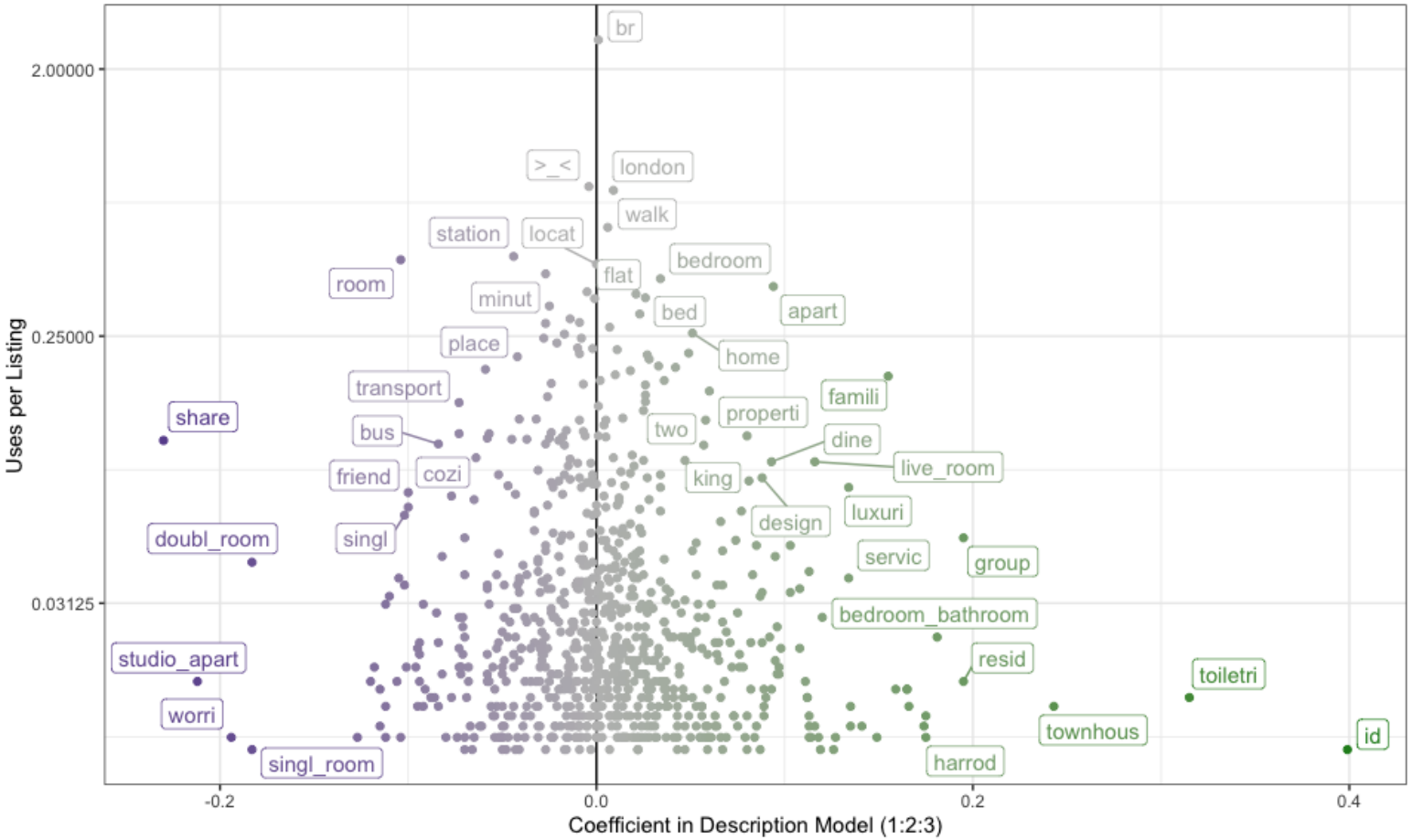
Economic Interpretation
Airbnb Pricing

High-price signals:
- luxury
- harrods
- townhouse
- serviced

Low-price signals:
- shared
- single_room
- Bus

Interpretation

Text captures:
- *Prestige*
- *Space*
- *Exclusivity*
- *Transportation access*



<u>Language reflects economic quality.</u>

Model Evaluation Strategy

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We use:	Train/Test split
	Stratification on log price
	Cross-validated LASSO
	RMSE & MAE
We compare to benchmarks to measure incremental value	
This follows best practices from class.	

Boundary Conditions

Airbnb Pricing

When might this fail?

- Extreme luxury listings
- Time-varying demand shocks
- Missing neighborhood income
- Nonlinear pricing mechanisms
- LASSO assumes linear effects.

Future improvements:

- Dynamic pricing effects
- Borough-level income merge
- Nonlinear models (tree-based)
- Review sentiment
- Central vs Outer London subsample

Conclusion
Airbnb Pricing

Titles contain limited signal

Descriptions contain meaningful economic information
Structured housing features dominate price
Text provides incremental predictive power
Transfer learning adds semantic richness

Final Takeaway:

Airbnb descriptions encode economically meaningful signals beyond observable housing characteristics.

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Thank you

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